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ORIGINAL ARTICLE

Efficacy of Action Observation Therapy on Patients with Dysphagia After Stroke: A Randomized Controlled Trial

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Abstract

To investigate the efficacy of Action Observation Therapy (AOT) which is conducted by observing and imitating swallowing videos for post-stroke dysphagia. 36 patients with post-stroke dysphagia were randomly assigned to an observation group or a control group equally. Both groups received routine dysphagia rehabilitation. AOT, was conducted for observation group while control group watched a same-duration landscape video before lunch and dinner. Before and after treatment, Watian Swallowing Test (WST), Eating Assessment Tool (EAT-10), Standard Swallowing Assessment (SSA), Functional Oral Intake Scale (FOIS), surface electromyogram (sEMG) and functional near-infrared spectroscopy (fNIRS) were conducted. After 3 weeks of treatment, WST, EAT-10, SSA, and duration of swallowing muscle groups of both groups significantly decreased ($P < 0.05$). FOIS and amplitude of swallowing muscle groups of both groups significantly increased ($P < 0.05$). Except for duration of subhyoid muscles and FOIS, other parameters in observation group were significantly improved compared to those in control group ($P < 0.05$). fNIRS showed that overall brain function connection was not significantly enhanced, but significant differences in bilateral primary motor cortex/supplementary motor cortex (PMC/SMC), left dorsolateral prefrontal cortex (DLPFC) and right Broca area were observed. Correlation analysis revealed that changes in amplitude of suprahyoid group muscle had a negative correlation with EAT-10 and SSA scores ($r = -0.332, -0.421$, $P < 0.05$). AOT based on observation and imitation of swallowing videos can enhance the strength of swallowing related muscles (such as the suprahyoid muscle group), shorten swallowing time, optimize the swallowing process, and ultimately promote the recovery of post-stroke swallowing function. Its mechanism may be related to the activation of brain regions such as PMC/SMC, DLPFC, and Broca area.

Keywords Action observation therapy · Surface electromyogram · Functional near-infrared spectroscopy · Dysphagia; stroke

Post-stroke dysphagia (PSD) is a common complication of stroke with a high incidence of 46.6% [1]. The prevalence of dysphagia in ischemic stroke and hemorrhagic stroke was 43.6% and 58.8%. The prevalence of dysarthria in PSD was 54.5%. The persistence of dysphagia at discharge and at 1 month was 74.5% and 50.9% respectively. Studies with large sample sizes provide robust evidence of their association with aspiration pneumonia, with odds ratios (ORs) ranging from 3.0 to 7.3 [2, 3]. An increased prevalence of dysphagia will result in an increased healthcare burden, both from resources spent on treating dysphagia and complications stemming from undiagnosed and thus untreated dysphagia.

Current treatment programs for PSD primarily involve indirect swallowing training (oropharyngeal muscle exercises), direct strategies (feeding training), compensatory strategies, and other supplementary measures such as swallowing neuromuscular electrical stimulation [4]. Numerous studies have shown that the earlier the rehabilitation starts, the better the prognosis patients get. For stroke patients,

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rehabilitation is important. For PSD, swallowing training is necessary and effective [5].However, majority of treatments need to be conducted by professionals within specialized medical institutions, making it challenging to extend the care to community and family. Therefore, it is imperative to continue exploring other feasible and effective methods for swallowing rehabilitation.

AOT is a new rehabilitation strategy based on the mirror neuron system (MNS), which uses videos or other forms of learning to guide patients to observe and imitate specific movements to promote neurological remodeling. Research indicates that AOT results in favorable therapeutic outcomes in the rehabilitation of upper and lower limb functions as well as aphasia post stroke [6, 7]. It is worth noting that the neuromodulation mechanism of swallowing may be closely related to MNS, as it is a partially observable action. Ushioda [8] is covered that BA6 and BA40 regions, which belongs to mirror neuron system, are activated while observing the action of drinking water. This indicates the existence of swallowing-associated MNS in the human brain, providing a theoretical basis for AOT to improve swallowing function of PSD. Compared to other traditional rehabilitation approaches, AOT has distinct advantages. Firstly, AOT can be applied at any stage of stroke, independent of patients' residual motor functions. Secondly, it requires no special equipment or dedicated facilities, has low costs, and is highly safe and easy to operate. AOT guides patients to actively participate in the training only through video media, which can be carried out continuously in the home nursing. Not only does it prolong the duration of clinical treatment and consolidate gains in function recovery but also reduces the occurrence of complications such as pneumonia, thereby enhancing the overall quality of life. However, there are few randomized controlled trials (RCTs) for this therapy in context of dysphagia [9–14]. Additionally, the mechanisms of how AOT influences the corresponding swallowing functional areas in the brain are not entirely clear at present, and further studies are still needed.

In this study, we proposed a multimodal evaluation method combing sEMG with fNIRS to investigate systematically the efficacy of AOT in PSD and the mechanism of brain function remodeling. SEMG is an objective indicator of the swallowing motor pathway from bottom to top, reflecting the collection of activity from the effector to the swallowing center, by detecting the electrical signals of swallowing muscle groups during the pharyngeal phase, because swallowing disorders arising from different causes exhibit distinctive patterns of sEMG signals in terms of both frequency and amplitude. It is an objective indicator of the swallowing motor pathway from bottom to top, reflecting the collection of activity from the effector to the swallowing center. FNIRS is a non-invasive brain imaging technique, which can reflect the "top-down" neuromodulation network reorganization pattern, by monitoring the cerebral hemodynamic changes during the swallowing task [13]. Unlike other neuroimaging techniques, such as functional magnetic resonance imaging(fMRI) and single-photon emission computed tomography (SPECT), fNIRS collects data by wearing a cerebral oximeter monitor on the head directly so it is less affected by participants' movement in a certain collection environment [14]. It also has the advantages of low sensitivity to motion artifacts and high temporal resolution [15].It has also been demonstrated that there was a good temporal and spatial correlation between fNIRS and fMRI activation during simultaneous multimodal recording of motor tasks [16]. As a result, fNIRS is increasingly being used in neuroimaging studies [17].

This research aims to verify the following scientific hypotheses: AOT may activate swallowing-related MNS networks and enhance peripheral muscle synergy and central nervous system plasticity simultaneously, thereby improving swallowing function in patients with PSD. The results of this study can provide evidence-based medical basis for promotion of application of AOT in swallowing rehabilitation, and new ideas for neuromodulation treatment of swallowing disorders.

Materials and Methods

General Information

From December 2023 to June 2024, 36 patients admitted to the Rehabilitation Department were selected as research subjects. Inclusion criteria were as follows: (1) meeting the diagnostic criteria of the "Chinese Guidelines for the Diagnosis and Treatment of Cerebral Hemorrhage (2019) " or the "Chinese Guidelines for the Diagnosis and Treatment of Acute Ischemic Stroke (2018)" confirmed by imaging examinations; (2) having varying degrees of dysphagia, with the Watada Drinking Water Test ranging from level 2 to level 5; (3) first onset of stroke and within a disease course of less than 6 months; (4) stable vital signs, cognitive ability, and the ability to understand and follow simple instructions; (5) binocular visual acuity or corrected visual acuity not less than 1.0; hearing test less than or equal to 25 decibels; (6) informed and voluntary participation by both patients and their families.

Exclusion criteria included: (1) occurrence of stroke more than once; (2) dysphagia caused by other reasons, such as motor neuron disease, postoperative complications of head and neck tumors, etc.; (3) presence of visual field defects or hemispatial neglect; (4) severe mental disorders

or cognitive impairments that hinder cooperation; (5) other conditions preventing completion of the experiment.

Dropout criteria were: (1) occurrence of severe complications or worsening conditions requiring rescue; (2) discharge before completing the intervention period; (3) poor compliance leading to self-withdrawal from the trial.

The study was approved by the Medical Ethics Committee of Sichuan Provincial People's Hospital (ChiCTR2400085845). According to a random number table, the patients were divided into an observation group and a control group, consisting of 18 cases in each.

Methods

Both groups received routine medications, conventional rehabilitation training, and routine swallowing function training. Depending on each patient's specific conditions, medications, were administered along with corresponding functional training like movement therapy, occupational therapy, cognitive training and speech training except swallowing function training. Routine swallowing function training included oral and facial muscle exercises, breathing exercises, pharyngeal ice stimulation, feeding training, swallowing posture training and Mendelsohn maneuver. Each session of routine training lasted 20 min, once a day, five days a week, for three consecutive weeks.

AOT utilizes an internal swallowing video consisting of three main modules: food stimulation, swallowing image demonstration, and feeding guidance. This video features a rich and colorful food montage, an animated breakdown of swallowing mechanisms, and a feeding scene with synchronized swallowing sounds. The 3.5-minute food stimulation section showcases various appetizers accompanied by cheerful background music, aiming to immediately capture the attention of participants. Next is a 2.5-minute animated tutorial that explains normal swallowing physiology, including oral chewing, tongue manipulation, oropharyngeal dynamics, and transport from the esophagus to the stomach. Finally, a 4-minute food swallowing module showcased a video of the demonstrator eating (front and side views), emphasizing chewing movements, pharyngeal contractions, and amplified swallowing acoustics to enhance multisensory observation.

AOT interventions are as follows. The completed swallowing action video was placed in front of the patient. Patients were instructed in advance to imitate swallowing actions shown in the video both during and after watching the video. The duration of watching was 10 min, and patients completed a 5-minute imitation training under the verbal guidance of a therapist. The control group watched an ordinary landscape video for 10 min, followed by a 5-minutes swallowing action training under verbal guidance from a different therapist. Verbal guidance means helping patients understand the stages of swallowing process and the key points of actions in each stage, guiding patients to swallow correctly to reduce the risk of aspiration. The whole training was conducted before lunch and dinner with each training lasting 15 min. Interventions were administered twice a day, five times a week, continuously for three weeks. As shown in Fig. 1.

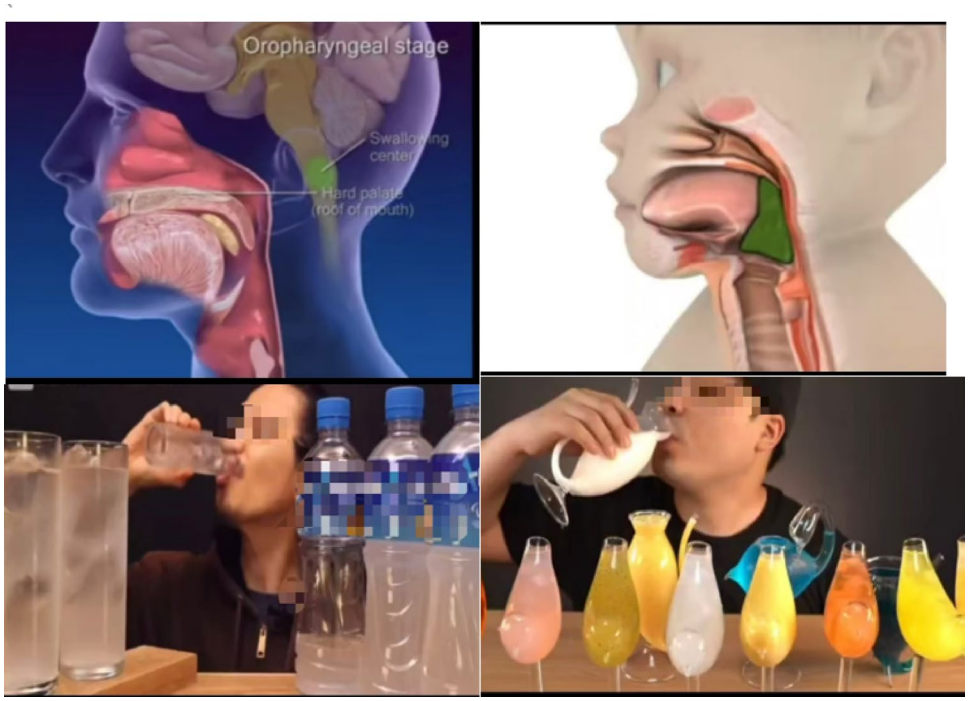

Fig. 1 AOT-based swallowing rehabilitation

Outcome Measures

WST, EAT-10, SSA, FOIS, sEMG, and fNIRS were completed by two groups before and after treatment.

WST [18]

The patient was given 30 mL of warm water in a sitting position. The drinking time and choking were observed. The outcome was divided into 5 grades with the highest grade indicating the worst swallowing function.

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EAT-10 scale [19]

10 questions associated with dysphagia help identify signs of aspiration and silent aspiration as well as signs of abnormal swallowing with higher scores representing worse swallowing function.

SSA scale [20]

The scale mainly includes three parts to assess swallowing funtion which includes general condition of patients, drinking 5 ml of water and drinking 60 ml of water. The total score ranges from 18 to 46 points with a higher score indicating worse swallowing function.

FOIS grading [21]

This 7-level scale is used to assess subjects' ability to eat by mouth with a higher grade indicates better oral feeding ability.

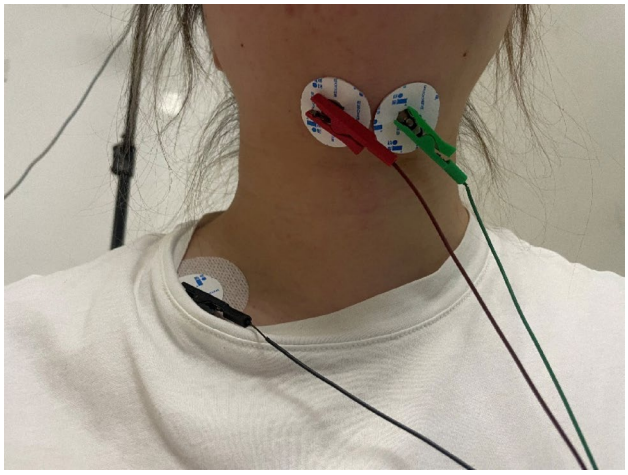

Fig. 2 sEMG electrode

SEMG

Surface Electromyography Instrument was manufactured by Shanghai Nuocheng (Model: NTS-2000). The skin was cleaned with 75% alcohol to ensure that baseline impedance values fell within the vicinity of 20 KΩ. The surface electrode for collecting electromyograph for the suprahyoid muscle group was located on skin and 1 cm above the thyroid cartilage in the midline of the neck. The recording electrode for the infrahyoid muscle group was placed 1 cm below the thyroid cartilage. The recording electrode(red) was placed on the affected side, the reference electrode(green) was placed on the countermeasure, and the ground wire(black) is connected to one electrode and placed on the bony bulge (clavicle). Patients were instructed to swallow air or saliva without substantial substance three times. The maximum amplitude and duration of swallowing muscle of electromyographic activity were recorded, and the average results were calculated. See Fig. 2.

FNIRS

Model BS-3000; Wuhan Znion Technology Co., Ltd. (Wuhan, China) was used for the test. The infrared wavelengths applied were 690 and 830 nm. There were 16 detecting and 16 emitting optrodes with 53 detection channels and a sampling rate of 20 Hz. Typical wearing method: prefrontal lobe。Using optical pole 9 as the FPz reference point, the left and right cerebral hemispheres are symmetrically distributed along the T3-T4 line with the hindbrain is fixed. See Fig. 3. The cerebral cortex was divided into different regions according to Brodmann's areas of brain with five regions for each hemisphere. The Channel diagram is Fig. 4. The relationship between channels and Brodmann area is shown in Table 1.

Examination process: firstly, personal particulars of patients, such as name and age, were entered. Signals were collected through a head-mounted device, and a quality check was conducted prior to data collection by checking the signal transmission rate of the optical probe. The test began when the transmission rate reached 90%. Patients were informed to remain calm for 30 s first. They were then instructed to keep their eyes closed, remain quiet at rest, maintain a peaceful mood, refrain from talking, avoid looking at phones, and refrain from engaging in activities with strong external stimuli, ensuring that brain was in a non-apparently active state. Then, the brain was continuously monitored in real time for 8 min during data collection. The brain function data were collected both before and after treatment.

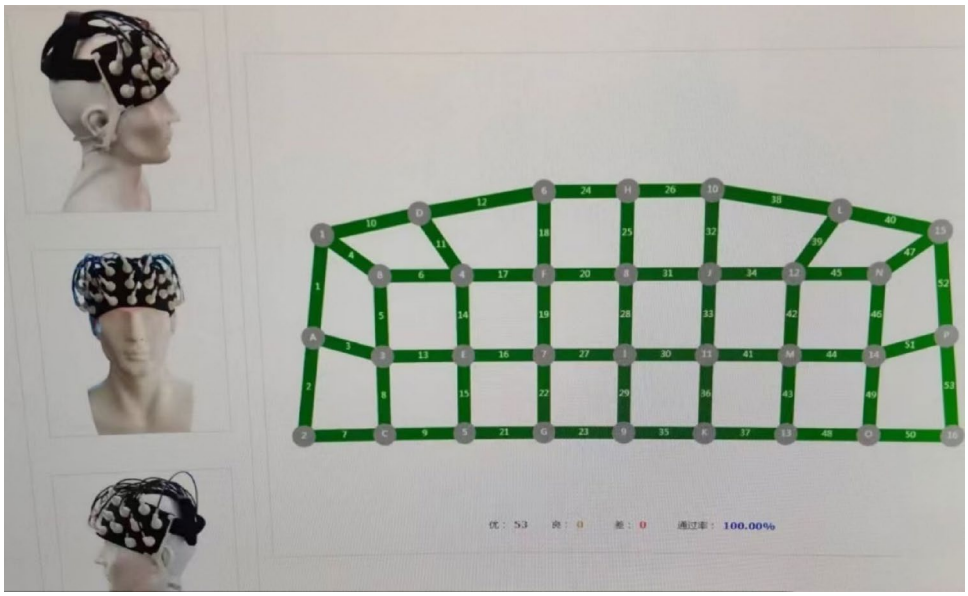

Fig. 3 Typical wearing method and transmission rate

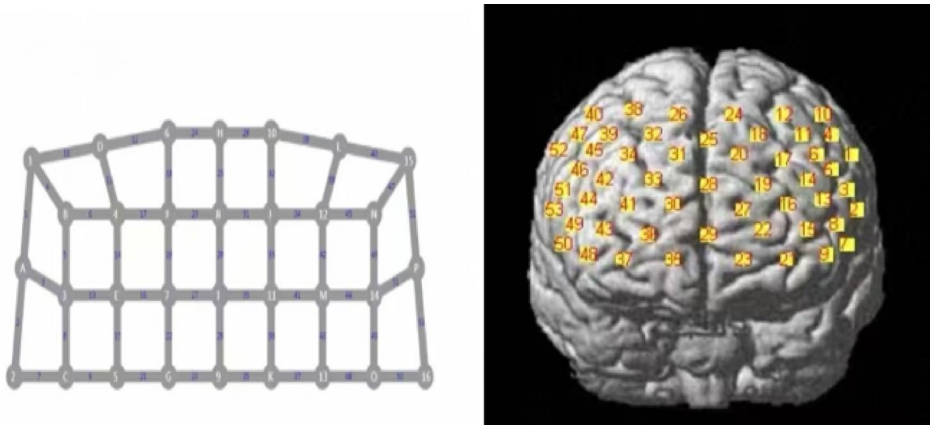

Fig. 4 Channel diagram

Table 1 Brodmann and channels

Brodmann area	Channels
PMC/SMC	CH01、CH04、CH10、CH40、CH47、CH52
Broca area	CH02、CH03、CH05、CH07、CH08、CH13、CH44、CH46、CH49、CH50、CH51、CH53
DLPFC	CH06、CH09、CH11、CH14、CH17、CH18、CH20、CH25、CH31、CH32、CH34、CH39、CH42、CH45、 CH48
Frontal eye field	CH12、CH24、CH26、CH38
Frontopolar area	CH15、CH16、CH19、CH21、CH22、CH23、CH27、CH28、CH29、CH30、CH33、CH35、CH36、CH37、CH41、CH43

Statistical Methods

Statistical analysis was conducted by using SPSS 26.0. Gender and stroke type were treated as categorical variables, and the chi-square test ($\chi 2$) was employed. Age, duration of illness, SSA, EAT-10, sEMG, and fNIRS data were treated as continuous variables and expressed as mean ± standard deviation. The t-test was used for comparison. Rank-sum tests were employed for inter-group comparisons of ordinal data. The significance level of $P<0.05$ was considered statistically significant.

Results

No significant difference was observed in gender, age, duration, and stroke type between the two groups ($P>0.05$), indicating comparability of the data (see Table 2; Fig. 5).

Swallowing Function

Compared to the pre-treatment values, WST, EAT-10 and SSA significantly decreased in both groups after treatment, and the observation group had lower values than the control group ($P<0.05$). After treatment, FOIS levels of the two groups were higher than that before treatment, and the differences between before and after intervention of

Table 2 Comparison of general data between two groups

	Observation group ($n=18$)	Contral group ($n=18$)	P value
Age (years, x(s))	67.17(11.47)	73.22(10.88)	0.113
Gender (M/F)	10/8	8/10	0.505
Stroke type (I/H)	14/4	15/3	0.674
Stroke onset to therapy (days)	19.17(18.36)	29.56(25.68)	0.172

For continuous data: mean (SD) or median (Q1; Q3).M, Male; F, Female; I, Infarction; H, Hemorrhage

two groups were also statistically significant ($P<0.05$), as shown in Table 3.

sEMG Results

No statistically significant difference between two groups was observed before treatment ($P>0.05$). After 3 weeks of training, sEMG amplitude of suprahyoid muscles and subhyoid muscles of both groups increased compared to those before treatment while the duration of swallowing muscles decreased ($P<0.05$). Except for the duration of subhyoid muscles ($P>0.05$), the range of changes in the amplitude of swallowing muscles and the duration of suprahyoid muscles in observation group were wider than those in control group, with a statistical significance ($P<0.05$), as shown in Table 4; Fig. 6. (1 means the observation group and 2 means the control group)

fNIRS Results

Data were extracted from 50 s to 230 s to ensure consistent data duration. Two data analysis approaches based on threshold values and sparsity were utilized to calculate graph theory metrics and conduct statistical analysis. Graph theory metrics included global and nodal metrics. Functional connectivity matrices for the two groups based on Brodmann's areas were generated, and bar graphs for significant differences were plotted. Statistical analysis on the number of connected edges under different thresholds was

Fig. 5 Flowchart

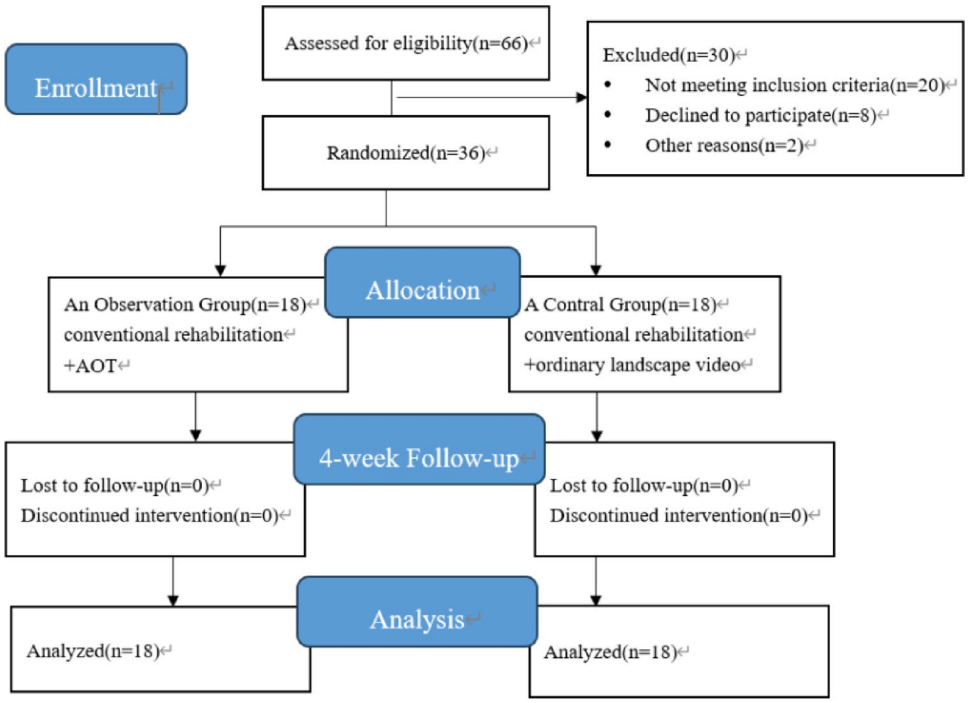

Table 3 Comparison of swallowing scale results

	Observation group(n=18)		Control group(n=18)		Differenceof observation	Differenceof control	pbetween group
	Before	After	Before	After			
WST(level)	3.78(1.22)	1.83(0.71)[c]	3.56(1.04)	2.17(0.86)[c]	−2.11(0.68)	−1.39(0.50)	0.002
FOIS(level)	2.61(1.46)	5.94(1.11)[c]	3.33(1.19)	4.78(1.17)[c]	1.44(0.51)	1.44(0.51)	1.000
EAT-10(score)	25.22(8.14)	7.33(4.77)[c]	24.00(7.81)	14.28(6.56)[c]	−17.89(5.27)	−9.72(2.99)	0.0001
SSA(score)	33.28(5.88)	21.44(3.88)[c]	31.78(6.21)	25.44(5.79)[c]	−11.83(4.41)	−6.39(2.17)	0.0001

(1) For continuous variables normally distributed pre-to-post evaluations with paired t-test ([a]$p<0.05$, [b]$p<0.01$, [c]$p<0.001$) and between-group comparisons using unpaired t-test; (2) for continuous variables not normally distributed: pre-to-post evaluations with Wilcoxon signed-rank test ([a]$p<0.05$, [b]$p<0.01$, [c]$p<0.001$) and between-groups comparison with the Mann-Whitney U test.

Table 4 Comparison of sEMG results between two groups

		Observation group ($n=18$)		Control group ($n=18$)		Difference observation	Difference control	p Between group
		Before	After	Before	After			
Suprahyoid muscles	Amplitude(mV)	0.265(0.049)	0.421(0.072)[c]	0.295(0.047)	0.377(0.044)[c]	0.155(0.066)	0.082(0.033)	0.0001
	Duration(s)	1.78(0.45)	1.09(0.25)[c]	1.71(0.30)	1.30(0.16)[c]	−0.69(0.34)	−0.40(0.23)	0.006
Subhyoid muscles	Amplitude(mV)	0.220(0.048)	0.346(0.048)[c]	0.234(0.043)	0.327(0.041)[c]	0.126(0.050)	0.092(0.026)	0.017
	Duration (s)	1.56(0.22)	1.07(0.17)[c]	1.53(0.22)	1.15(0.14)[c]	−0.49(0.19)	−0.31(0.19)	0.087

For continuous variables normally distributed pre-to-post evaluations with paired t-test ([a]$p<0.05$, [b]$p<0.01$, [c]$p<0.001$) and between-group comparisons using unpaired t-test.

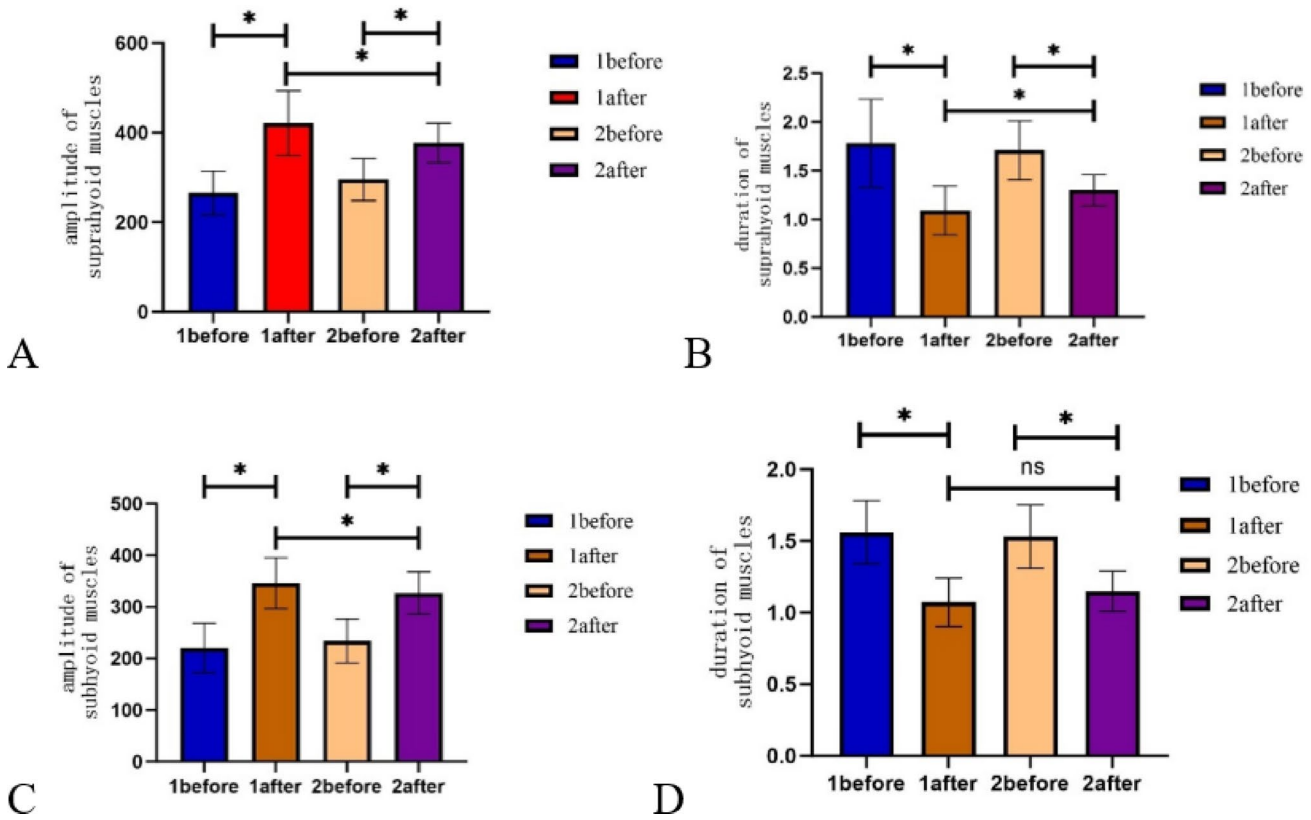

Fig. 6 sEMG results

conducted. Topographical maps based on channels and ROI levels were created. In the Figures, 1 represents the observation group, and 2 represents the control group.

Global Metrics

The global metrics consisted of five parameters, including the small-world attributes (aSigma), clustering coefficient (aCP), shortest path length (aLp), global efficiency (aEg), and local efficiency (aEloc). These five global metrics were individually calculated, and their within-group main effects, between-group main effects, interactions, and post-hoc tests showed no significant difference($P>0.05$).

Functional Connectivity Matrices

Two sets of brain functional connectivity matrices were generated for the two groups before and after treatment. Various colors were used to represent different levels of connectivity strength, with x and y-axes representing different brain regions on the left and right hemisphere. Pairwise sample t-tests were conducted for the observation group before and after treatment, as well as for the control group before and after treatment. Additionally, two-sample t-tests were performed for pre-treatment and post-treatment data of both groups. After multiple comparison correction (FDR), no significant differences were found, as depicted in Fig. 7.

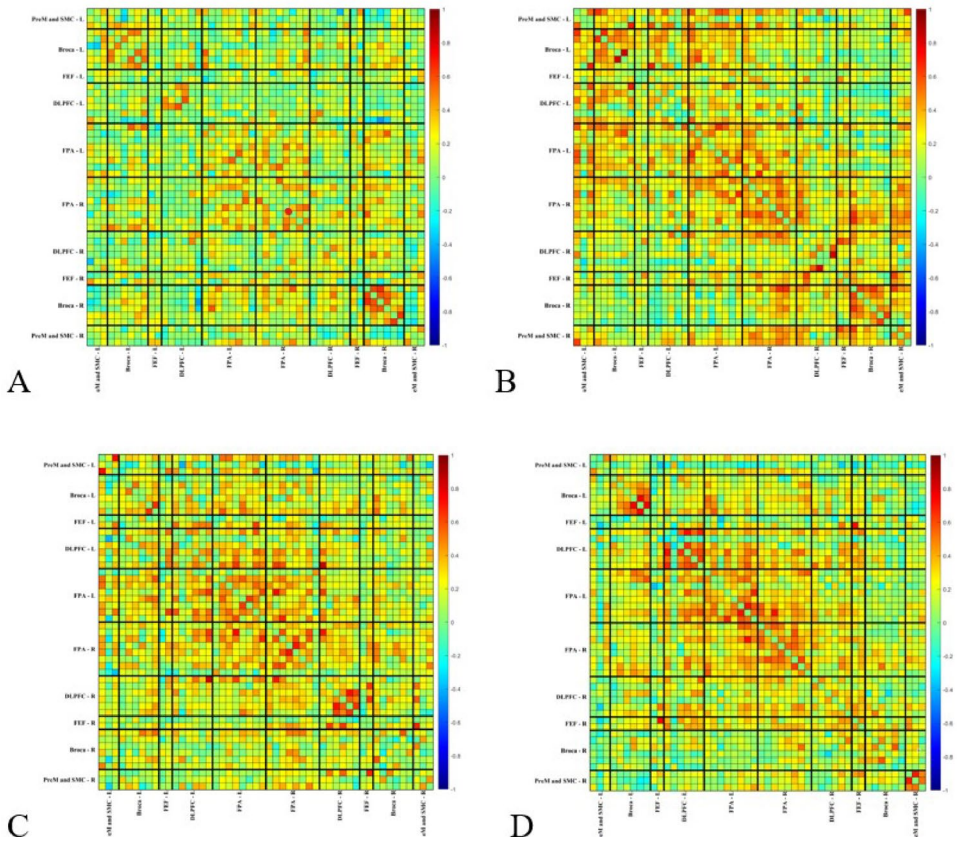

Fig. 7 Functional connectivity matrices. （**A**: before treatment of observation group **B**: after treatment of observation group **C**: before treatment of control group **D**: after treatment of control group）

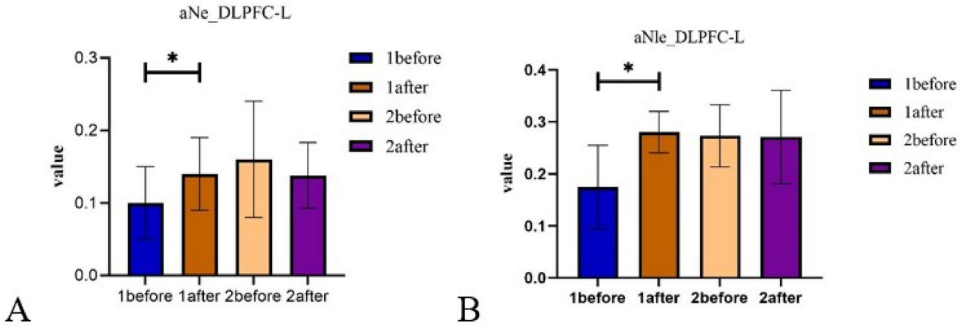

Fig. 8 DLPFC-L：aNe, aNLe

Node Metrics

Node metrics, which are indicators reflecting the importance of a specific local region in the brain, include betweenness centrality (aBc), degree centrality (aDc), node clustering coefficient (aNcp), node efficiency (aNe), and node local efficiency (aNLe). Brain regions with statistically significant differences include Broca, DLPFC, and PMC/SMC (PreM_and_SMC in figures). Refer to Figs. 8, 9 and 10. In the figures, R represents the right side of brain, L represents the left side of the brain.

Correlation of Swallowing Scale and Surface Electromyography

Correlation tests were conducted using the differences in swallowing scale scores and sEMG data before and after treatment for two groups. The Spearman correlation analysis revealed a negative correlation between the amplitude of the suprahyoid muscle group for both the EAT-10 and SSA scales ($r = -0.332, -0.421$, $P < 0.05$). (See Fig. 11).

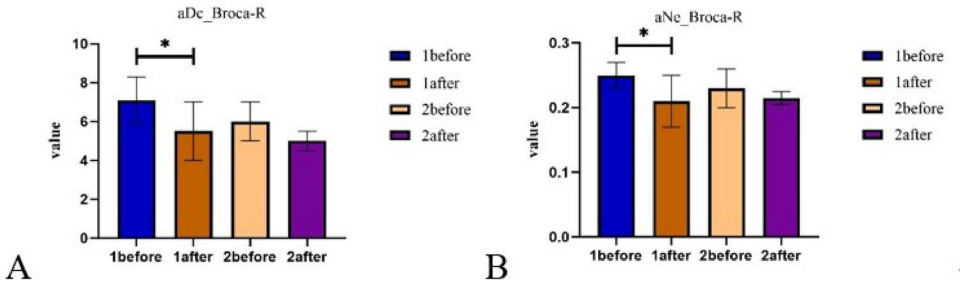

Fig. 9 Broca-R：aDc, aNe

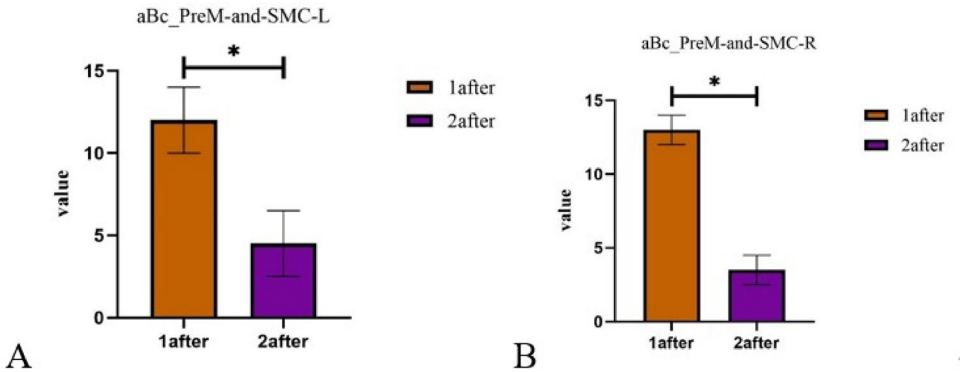

Fig. 10 aBc: PreM_and_SMC-L, PreM_and_SMC-R

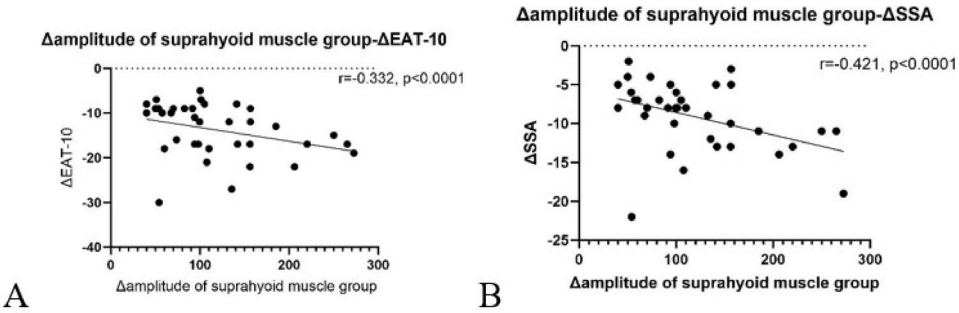

Fig. 11 Correlation between EAT-10 and SSA and amplitude of the suprahyoid muscle group

Discussion

The results of this study indicated that both groups showed improvement in swallowing scale scores and sEMG indicators of suprahyoid and infrahyoid muscle groups before and after treatment, suggesting that both treatment plans were effective. Comparing the results between two groups, the observation group showed statistically significant improvements in EAT-10, SSA, FOIS, the amplitude of suprahyoid and infrahyoid muscle groups, and the duration of suprahyoid muscle groups. Combining sEMG indicators with swallowing scales, amplitude of suprahyoid muscle group and EAT-10 scales had a negative correlation, and so does amplitude of suprahyoid muscle group and SSA scores.

Based on these results, it is hypothesized that AOT can enhance the muscle strength of major swallowing muscle groups, shorten swallowing duration, and thereby improve swallowing function in post-stroke patients. From a brain functional perspective based on fNIRS results, specific brain regions such as bilateral PMC/SMC, left DLPFC, and right Broca's area showed significant difference, although there was no apparent enhancement in overall brain functional connectivity.

Cortical areas associated with swallowing primarily include the SMC, insular cortex, PMC, cingulate cortex, and PFC. Knollhoff et al. [22]reported that bilateral PMC, DLPFC, inferior frontal gyrus (IFG), supplementary motor area (SMA), and primary somatosensory cortex (PSC) were all activated during swallowing in healthy adults. The results of the fNIRS showed that bilateral PMC/SMC, left DLPFC, and right Broca area were activated during the patients' performance of AOT, which is highly overlapping with the swallowing center, MNS and the activation of brain regions of healthy people when swallowing [23]. The results of our study revealed that the values of aNe and aNle in left DLPFC in observation group were increased in comparison with that before treatment, whereas no similar changes were observed in control group. The DLPFC

belongs to the ventral mirror system and has extensive connections with other brain regions [24]. It involves in various cognitive control tasks including working memory, cognitive procedural processes, action planning, imitation, and abstract reasoning [25]. We believe that patients store the action mechanism in brains as motor memory through observation and understanding while watching videos of swallowing actions. With the assistance of MNS in perceiving, understanding, and imitating motor skills, the behavior of the operator is linked to the observer, forming the basis of reproducing swallowing action [26].

The study also found that after treatment, the values of aDC and aNe in the right Broca's area of observation group decreased compared to those before treatment, indicating a reduced centrality in this area. This finding contradicts the conclusions of earlier works which showed that cortical activation during swallowing is multifocal, bilateral, and independent of hemispheric dominance [27]. We think that this phenomenon could be explained by the theory of callosal inhibition. After a stroke, the competition and inhibition relationship between two brain hemispheres is disrupted, leading to excessive activation of the healthy hemisphere. After treatment, this excessive activation is suppressed, resulting in a decrease in centrality in right Broca's area. However, our study did not find a statistically significant increase in centrality in left Broca's area, and we hope that future researches could provide more insights to support this speculation.

Our study showed results consistent with those obtained earlier by Jing YH and others [28], who used task-based fMRI to study activation of brain regions, including the supplementary motor area, during the observation of swallowing actions. In our study, the aBC values of PMC/SMC on both sides in the observation group showed a statistically significant increase compared with the control group after treatment. We believe that this result may be related to the activation of mirror neuron system caused by observation and learning of similar swallowing movements. PMC/SMC belong to the dorsal mirror system and is a critical region of motor cortex, responsible for integrating information from different neural circuits and managing information related to motor programs such as self-initiated movements, movement sequences and motor learning [29]– [30]. During motor observation, the entire process comprising observation, learning, and imitation selectively activates major motor areas such as supplementary motor area, thereby regulating the swallowing function of patients.

In summary, AOT for swallowing is a neurorehabilitation method that demonstrates effects spanning central neural mechanisms to effector pathways. Its therapeutic mechanism likely involves selective activation of MNS. Through visual-auditory stimulation of mirror neurons, AOT engages both dorsal and ventral mirror pathways, thereby modulating motor, sensory, and cognitive neural networks. This neuroplastic adaptation facilitates reorganization of cortical swallowing circuits, enhancing functional brain remodeling and subsequent recovery of swallowing abilities. Thus, this study demonstrates that AOT utilizing institutionally developed swallowing video protocols-as a top-down therapeutic strategy-achieves clinical efficacy in post-stroke dysphagia rehabilitation. Furthermore, AOT requires no specialized equipment which can simplify the implementation process to enhance patients' engagement and reduce caregiver burden. AOT demonstrates cost-effectiveness while ensuring treatment continuity. It offers innovative strategies for home-based and tele-rehabilitation, thereby supporting a three-tier hospital-community-home integrated rehabilitation model for post-stroke dysphagia management.

Limitations and Expectation

This study has some limitations that entail future exploration:1) Short Intervention Period: The intervention duration was relatively short. Long Intervention Period could dynamically observe changes in activated brain regions. 2) Small Sample Size: The study included a limited number of participants.

We would like to see that future research could benefit from larger, multicenter, and randomized controlled trials. Real-time monitoring of brain function changes during motion observation therapy could deepen our understanding of the AOT mechanism. Combining AOT with other treatments may provide a comprehensive approach for sustained treatment.

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Data Availability Data are available on reasonable request.

Declarations

Competing Interest The authors have no conflict of interest to disclose related to this manuscript.

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